

User Documentation

Product: SoilSidekick Pro / LeafEngines

Version: 3.2.0

Last updated: 2026-07-10

1. Who This Is For

This guide is for **end users** of the SoilSidekick Pro web app and consumer integrations. Developer-focused material lives in the Developer Specifications and API Documentation.

2. What's New in 3.2.0

- **New consumer tier naming** — Hobby (\$29), Grower (\$79), **Pro (\$149)** — replacing legacy Basic/Standard/Premium labels. Manage at [/plans](#) (SSP consumer). The [/pricing](#) page remains the LeafEngines API tier page.
- **Managed Assets API (SDK v3.1.0)** — full CRUD for fields, sensors, and points via [assets-crud](#), with optimistic locking ([If-Match](#)), audit history, and soft delete.
- **WFS Export** — [wfs-export](#) edge function exposes your assets as an OGC WFS layer for QGIS/ArcGIS.
- **Dual-meter API keys** — separate free-tier and paid quotas on the same key, with [pg_cron](#) monthly resets.
- **QGIS Plugin v1.0.10** — WFS-T fixes, Bandit B411 security hardening, improved layer styling. Install via QGIS Plugin Manager (ID 4987).
- **Telegram Bot** — link your account with a one-time code at [@LeafEnginesBot](#) and run soil/plant queries from chat.
- **SoilCertify GIS resources** — infrastructure use case, developer landing, and technical overview cross-linked under [/docs/workflows/soilcertify/](#).
- **FAQ refreshed** (2026-11-19) with current pricing and workflows.
- **Security hardening** — RLS on [api_keys](#) and [model_benchmark_results](#), restricted [vendor_leads](#) to admins, secured [telegram-uploads](#) bucket.

3. Getting Started

3.1 Try Without Signup (Free Tier)

- Visit [soilcertify.com](#) or [app.soilsidekickpro.com](#)
- Use test key [leaf-test-370df0a2e62e](#) or click "Try Free"
- Enter a US county name or GPS coordinates
- View soil composition, pH, NPK, and recommendations

3.2 Create an Account

- Go to [app.soilsidekickpro.com/sign-up](#)

- Sign in with email or Google
- Complete profile — pick your role (Farmer / Consultant / Developer / Enterprise)
- Verify email via the branded welcome message

3.3 Choose a Plan

Audience	Route	Tiers
Consumer / SSP	/plans	Hobby \$29 · Grower \$79 · Pro \$149
Developers / LeafEngines API	/pricing	Free · Starter \$149 · Pro \$499 · Enterprise

Free features remain free forever. Paid plans unlock quotas, VRT, AI chat, satellite intelligence, and Managed Assets.

4. Core Workflows

4.1 Soil Analysis

- Dashboard → "New Analysis"
- Enter address, drop pin, or upload GPS trace
- Review NPK, drainage, texture, satellite overlay
- Export PDF or share link

4.2 Field Mapping ([/field-mapping](#))

- Draw polygons on the map
- Assign crop and season
- Save as a **Managed Asset** (v3.1.0+)
- Export as GeoJSON or WFS layer

4.3 Seasonal Planning ([/seasonal-planning](#))

- Auto-generated calendar using 5-method optimization
- Weather-adjusted planting windows
- Task templates for irrigation, fertilization

4.4 VRT Prescriptions

- Requires Pro or Enterprise
- Upload boundary or select field
- Generate variable-rate map
- Download as shapefile or ISOBUS-XML

4.5 AI Assistant

- Chat with GPT-5 grounded on your fields
- Falls back to on-device Gemma when offline
- Never shares your field data with 3rd parties

4.6 Offline Mode

- App works fully read-only offline
- Writes queue automatically and sync on reconnect
- If GPS is unreliable, the app blocks saves > 500 m uncertainty and shows a soft warning

5. Managed Assets (SDK v3.1.0+)

Your saved fields, sensors, and points are full assets:

- **Optimistic locking** via **If-Match** ETag prevents accidental overwrites
- **Audit history** in **asset_history** captures every change
- **Soft delete** keeps records recoverable for 30 days
- **Public toggle** (**is_public**) exposes selected assets via WFS
- **Rate-limited** per API key with dual-meter quotas

Manage them at **Dashboard** → **Assets**, or via **POST/GET/PATCH/DELETE /assets-crud**.

6. API Keys (for your own tools)

- app.soilsidekickpro.com/api-keys
- Click "New key" → name it → copy once (SHA-256 hashed; not recoverable)
- Add to your tool as **x-api-key** header
- Rotate anytime; revoke instantly
- Every key now tracks **free-tier** and **paid** usage independently

7. Integrations

- **QGIS Plugin v1.0.10** — install from QGIS Plugin Manager (ID 4987). Now includes WFS layer support and Bandit B411 hardening.
- **Claude Desktop (MCP)** — `npm i -g @ancientwhispers54/leafengines-mcp-server` — 10 tools including `get_soil_data`, `planting_optimization`, `carbon_credit_estimate`.
- **n8n** — `npm i n8n-nodes-leafengines`
- **Node-RED** — `npm i node-red-contrib-leafengines`
- **Telegram bot** — `/start` at [@LeafEnginesBot](#), link account with one-time code
- **SDKs (v3.1.0)** — TypeScript, Python, Ruby, PHP, Java, Go — all published to their respective registries with Asset Management and WFS export APIs

8. Consumer Plant Care

- Take a photo → get safe identification
- Care schedule adapts to your local FIPS regulations
- Chemical warnings tuned to your jurisdiction
- Progressive-jargon toggle for beginner or expert view

9. Notifications

- Push notifications for tasks, alerts, sensor anomalies
- Email digest weekly
- Haptic feedback on supported devices

10. Privacy & Data Ownership

- You own your data
- PII encrypted (AES-256 V3) at rest
- Local AI never sends your queries to the cloud
- Export or delete your account any time from Settings

11. Troubleshooting

Symptom	Fix
"Position uncertain" warning	Move outdoors, wait 10 s for GPS fix
Offline sync stuck	Reopen app while online; queue drains automatically
API key rejected	Confirm you're on app. subdomain, not web.
AI response missing	Check internet or enable local mode in Settings
Missing satellite overlay	Requires Pro tier
Asset update returns 412	Refetch the asset for the latest If-Match ETag and retry
WFS layer empty in QGIS	Verify assets are marked is_public=true or the key is authorized

12. Accessibility

- WCAG 2.1 AA
- Keyboard navigable throughout
- High-contrast theme in Settings

- Screen-reader tested (VoiceOver, NVDA)

13. Language & Regions

- English (default)
- Localized units (imperial/metric) per profile
- Regional pricing shown in local currency (US, EU, UK, AU)

14. Support Channels

- In-app chat (business hours)
- Email support@soilsidekickpro.com
- Community: github.com/Qwarranto/soil-sidekick-pro-guide/discussions
- Enterprise SLA phone line (Enterprise tier)

15. Related Guides

- Workflow guides: [docs/workflows/](#)
- SoilCertify GIS: [docs/workflows/soilcertify/](#)
- QGIS developer landing: [docs/workflows/15_QGIS_DEVELOPER_LANDING.md](#)
- Quick start: [docs/get-started/QUICK_START.md](#)
- Telegram: [docs/telegram/TELEGRAM_QUICK_START.md](#)
- SDK reference: [docs/sdk/DEVELOPER_SPECIFICATIONS.md](#)